Spatial dimension compactification in general relativity and duality symmetry

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Notations and abreviations

For the abreviations we wil use

- GR for general relativity

We will during this intership use the vielbein formalism. We will use the same notation as in [**scherk1979get**] which is

- Greek hatted letter for curved indices $(\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots)$
- Latin hatted letter for flat indices $(\hat{p}, \hat{m}, \hat{n}, \dots)$
- For space-time indices taking D values, late greek letters are used in the curved case $(\mu, \nu, \rho, \dots)$ and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case $(\alpha, \beta, \gamma, \dots)$ and Latin letters in the flat case (a,b,c)

THe general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α}. The flat lorenz

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1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [1] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [2] and later refined by Klein:
by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [3] and string theory.

2 General relativity theoretical reminder

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

2.1 mathematical basis

Definition 1 *Let V be an R-vector space of dimension N, we define the tensor of order* $(r, s) \in [\![0, 1]\!]^2$ *on* $V^{(r,s)} \overset{def}{=} V^{\otimes r} \otimes V^{\otimes s}$ *as a being a multilinear application on* $V^{\star r} \times V^s$ *to* $\mathbb{R}$

Definition 2 *We will call n-dimension* ($n \in \mathbb{N}^*$) *manifold every Hausdorff topological space* $(\mathcal{M}, \tau)$ *localy homeomorphic to* $\mathbb{R}^n$)

Definition 3 *Let* $(\mathcal{M}, g)$ *be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form* $g_{\mu\nu}$ *on the tangent space* $T_p\mathcal{M}$ *at each point* $p \in \mathcal{M}$, *such that*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 4 *In general in RG we use the christofells connexion as it is the sole one wich is mtreci and torsion free. We define it as being*

$$nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\sigma} V^\sigma \tag{1}$$

where the christofells symbols are

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \tag{2}$$

2.2 spin connexion, vielbein and flatspace

3 Lagrangian field theory

3.1 Lagrangian density

In field theory we will rewrite the lagrangian in this way

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz \tag{3}$$

where $\mathcal{L}$ is the lagrangian density, but we will simply refer to it as the lagrangian.
And so we can no write the action as being

$$S = \int \mathcal{L} dx^\mu \tag{4}$$

3.2 General exemple for scalar and vectorial fields

There is classical forumlation for scalar and vectorial field and we will use their version without source as there are not that interssting in our endavour for vectorial field we have

$$\boxed{\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi} \tag{5}$$

and for a vectorial field we have

$$\boxed{\mathcal{L} = -\frac{1}{-}F_{\mu\nu}F^{\mu\nu}} \tag{6}$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A^\nu - \partial_\nu A^\mu$$

now in the futur we also need the lagrangian for RG wich will simply be einstein action,

$$\boxed{\mathcal{L} = R} \tag{7}$$

where R is ricci scalar and the lagrangian of a two form fields which is given in [4] by

$$\boxed{\mathcal{L} = g^{\hat{\mu}\mu'}g^{\nu\nu'}g^{\rho\rho'}H_{\mu\nu\rho}H_{\mu'\nu'\rho'}} \tag{8}$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

4 Reduction of hilbert-einstein Action in D+E dimension

Here I will do a kalusa klein reduction. Basically we will suppose an universe in D+E dimension where D is 4 and we will try to see how on observater in D will perceive the new dimension.

4.1 Hypothesis

We will consider that the fields don't depend on the internal variable of the world so for any field $\partial_\alpha A^{\hat{\mu}} = 0$

4.2 Rewriting the action

4.3 vielbein anstazs and latin connexion

4.4 finalisation

5 reduction of the new fields

5.1 scalar field reduction

5.2 Two form fiel reduction

6 Observation of new symmetry

7 Conclusion

References

[1] Albert Einstein. "Zur allgemeinen relativitätstheorie". In: *Sitzungsber. preuss. Akad. Wiss* 44.2 (1915), pp. 778–786.

[2] Th Kaluza. "Zum unitätsproblem der physik". In: *Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.)* 1921.arXiv: 1803.08616 (1921), pp. 966–972.

[3] Daniel Z Freedman and Antoine Van Proeyen. *Supergravity*. Cambridge university press, 2012.

[4] Jnanadeva Maharana and John H Schwarz. "Noncompact symmetries in string theory". In: *Nuclear Physics B* 390.1 (1993), pp. 3–32.

A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}
\delta S &= \delta \int \mathcal{L} dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho}\delta\eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\partial_\nu\eta_\rho dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho}\delta\eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\eta_\rho + \underbrace{\partial_\nu\left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\eta_\rho\right)}_{\text{border terme equal to zero}} \quad dx^\mu \\
&= 0
\end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \tag{9}$$

which is in fact the equation of mouvement for the field